United States Patent	12402782
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Christensen; Martin Johst et al.

Endoscope control system

Abstract

An endoscope including a control system including: a first control wheel connected to a first wire drum and to a first steering wire of the endoscope to control the bending operation in a first dimension, the first control wheel having a bearing surface; a second control wheel connected to a second wire drum and to a second steering wire to control the bending operation in a second dimension, the second control wheel having an outer bearing surface positioned farther from the axis of rotation than the bearing surface of the first control wheel; and an outer bearing element including an inner bearing surface positioned farther from the axis of rotation than the outer bearing surface of the second control wheel and abutting the outer bearing surface of the second control wheel so that rotation of the second control wheel is at least partly borne on the outer bearing element.

Inventors: **Christensen; Martin Johst (Copenhagen, DK), Schütz; Günter Wilhelm (Augsburg, DE)**

Applicant: **AMBU A/S** (Ballerup, DK)

Family ID: **78291995**

Assignee: **AMBU A/S (Ballerup, DK)**

Appl. No.: **17/239373**

Filed: **April 23, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20210338050 A1	Nov. 04, 2021

Foreign Application Priority Data

EP	20172237	Apr. 30, 2020
EP	20172238	Apr. 30, 2020

Publication Classification

Int. Cl.: **A61B1/00** (20060101); **A61B1/005** (20060101); **B29C65/58** (20060101); B29L31/00 (20060101)

U.S. Cl.:

CPC **A61B1/0052** (20130101); **A61B1/0011** (20130101); **A61B1/0057** (20130101); **B29C65/58** (20130101); B29L2031/753 (20130101)

Field of Classification Search

CPC: A61B (1/0052); A61B (1/0011); A61B (1/0057); A61B (1/00105); A61B (1/00066); G05G (1/12); F16B (21/082); F16B (21/086); F16B (21/084); F16B (21/088)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3610231	12/1970	Takahashi	N/A	N/A
4012155	12/1976	Morris	403/360	A47B 87/0207
4207873	12/1979	Kruy	N/A	N/A
4461282	12/1983	Ouchi et al.	N/A	N/A
4473301	12/1983	Namyslo	N/A	N/A
4617914	12/1985	Ueda	N/A	N/A
4825850	12/1988	Opie et al.	N/A	N/A
4924852	12/1989	Suzuki et al.	N/A	N/A
4942866	12/1989	Usami	N/A	N/A
5014685	12/1990	Takahashi	N/A	N/A
5086200	12/1991	Kline	200/11R	H01H 19/115
5174276	12/1991	Crockard	N/A	N/A
5269202	12/1992	Kiyosawa et al.	N/A	N/A
5329887	12/1993	Ailinger et al.	N/A	N/A
5426992	12/1994	Morii et al.	N/A	N/A
5464007	12/1994	Krauter et al.	N/A	N/A
5507717	12/1995	Kura et al.	N/A	N/A
5512035	12/1995	Konstorum et al.	N/A	N/A
5575755	12/1995	Krauter et al.	N/A	N/A
5871441	12/1998	Ishiguro	600/159	A61B 1/122
5888192	12/1998	Heimberger	N/A	N/A
6288351	12/2000	Bruntz	200/11R	H01H 3/10
6599265	12/2002	Bon	N/A	N/A
6656111	12/2002	Fujii et al.	N/A	N/A
6673012	12/2003	Fujii et al.	N/A	N/A
6821282	12/2003	Perry et al.	N/A	N/A
7510107	12/2008	Timm et al.	N/A	N/A
7588176	12/2008	Timm et al.	N/A	N/A
7633837	12/2008	Daout	N/A	N/A
7731072	12/2009	Timm et al.	N/A	N/A

7735396	12/2009	Ishikawa et al.	N/A	N/A
7926379	12/2010	Gutmann et al.	N/A	N/A
8042423	12/2010	Bannier et al.	N/A	N/A
8048025	12/2010	Barenboym et al.	N/A	N/A
8052636	12/2010	Moll et al.	N/A	N/A
8257303	12/2011	Moll et al.	N/A	N/A
8286845	12/2011	Perry et al.	N/A	N/A
8302507	12/2011	Kanai	N/A	N/A
8578808	12/2012	Koitabashi	N/A	N/A
8808168	12/2013	Ettwein et al.	N/A	N/A
8845521	12/2013	Maruyama	N/A	N/A
8904894	12/2013	Geiser	N/A	N/A
8911362	12/2013	Kaneko	N/A	N/A
9044135	12/2014	Ishii et al.	N/A	N/A
9044138	12/2014	Sjostrom et al.	N/A	N/A
9057421	12/2014	Ishikawa et al.	N/A	N/A
9155865	12/2014	Golden et al.	N/A	N/A
9237837	12/2015	Omoto et al.	N/A	N/A
9360098	12/2015	Roopnarine	N/A	N/A
9394985	12/2015	Kobayashi et al.	N/A	N/A
9457168	12/2015	Moll et al.	N/A	N/A
9534681	12/2016	Ishikawa	N/A	N/A
9545253	12/2016	Worrell et al.	N/A	N/A
9833131	12/2016	Golden et al.	N/A	N/A
9949623	12/2017	Lang et al.	N/A	N/A
10085623	12/2017	Osaki	N/A	N/A
10197153	12/2018	Dumanski et al.	N/A	N/A
10203022	12/2018	Atmur et al.	N/A	N/A
10238271	12/2018	Haraguchi	N/A	N/A
11786112	12/2022	Nielsen et al.	N/A	N/A
2001/0037051	12/2000	Fujii et al.	N/A	N/A
2002/0019591	12/2001	Bon	N/A	N/A
2002/0094232	12/2001	Lipp	403/329	F16D 1/108
2002/0099266	12/2001	Ogura et al.	N/A	N/A
2004/0010245	12/2003	Cerier	606/1	A61B 17/0057
2004/0015054	12/2003	Hino	600/146	G02B 23/2476
2009/0149709	12/2008	Koitabashi	N/A	N/A
2009/0247828	12/2008	Watanabe et al.	N/A	N/A
2011/0118550	12/2010	Tulley	N/A	N/A
2011/0144440	12/2010	Cropper	600/203	A61B 17/3421
2011/0208001	12/2010	Haeckl	600/125	A61B 1/0051
2012/0277535	12/2011	Hoshino	N/A	N/A
2013/0204096	12/2012	Ku et al.	N/A	N/A
2013/0296848	12/2012	Allen et al.	N/A	N/A
2014/0058323	12/2013	Hoshino	N/A	N/A
2014/0142389	12/2013	Lim et al.	N/A	N/A
2014/0296640	12/2013	Hoshino	N/A	N/A
2014/0343489	12/2013	Lang et al.	N/A	N/A
2015/0057537	12/2014	Dillon	600/113	A61B 1/0014
2015/0359415	12/2014	Lang	600/141	A61M 25/0133

2016/0067457	12/2015	Selkee	N/A	N/A
2018/0132899	12/2017	SooHoo	N/A	N/A
2019/0029498	12/2018	Mankowski et al.	N/A	N/A
2019/0035440	12/2018	Yuan et al.	N/A	N/A
2019/0209205	12/2018	Nishio	N/A	N/A
2019/0313884	12/2018	Isobe	N/A	N/A
2019/0350440	12/2018	Leong et al.	N/A	N/A
2021/0338049	12/2020	Christensen	N/A	N/A
2021/0338051	12/2020	Nielsen et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0306723	12/1992	EP	N/A
0754429	12/2003	EP	N/A
2594307	12/2012	EP	N/A
2692227	12/2013	EP	N/A
2692277	12/2013	EP	N/A
2594307	12/2015	EP	N/A
2692227	12/2017	EP	N/A
09-038028	12/1996	JP	N/A
2005-160790	12/2004	JP	N/A
2008/023965	12/2007	WO	N/A
2014/186519	12/2013	WO	N/A
2018/022402	12/2017	WO	N/A
2018/022418	12/2017	WO	N/A
2018/131305	12/2017	WO	N/A
2021/213600	12/2020	WO	N/A

OTHER PUBLICATIONS

EBay listing, steinelager.de, Wheel 32×64 Conical with Spikes and Inner 48 Tooth Gear 64712—Black, available at https://steinelager.de/en/category/7/wheel and https://www.steinelager.de/en/color/26/black, website copyright 2016-1029. cited by applicant

Extended European Search Report issued in EP20172237.8, mailed Oct. 19, 2020, 9 pages. cited by applicant

Extended European Search Report issued in EP20172238.6, mailed Oct. 26, 2020, 9 pages. cited by applicant

Extended European Search Report issued in EP20172242.8, mailed Oct. 12, 2020, 8 pages. cited by applicant

Industrial conical gear, cogwheel, available at Dreamstime.com, website copyright 2000-2019. cited by applicant

Jain et al., “Micromanipulator: Effectiveness in Minimally Invasive Neurosurgery,” Minim Invasive Neurosurg 2003; 46(4): 235-239. cited by applicant

Jarrahy et al., “A new powered endoscope holding arm for endoscopic surgery of the cranial base,” Minim Invasive Neurosurg 2002, 45(3): 189-192. cited by applicant

Lerner, “A Passive Seven Degree of Freedom Postitioning Device for Surgical Robots and Devices,” dissertation submitted to the Johns Hopkins University, Baltimore, Maryland, 1998. cited by applicant

Poels, Design of the Frame and Arms of a Master for Robotic Surgery, Traineeship report, Technische Universiteit Eindhoven, Department of Mechanical Engineer, Control Systems Technology Group, Jul. 2007. cited by applicant

Office Action in related U.S. Appl. No. 17/241,882 dated Dec. 23, 2022, 12 pages. cited by applicant
Final Office Action received for U.S. Appl. No. 17/239,372 dated Nov. 30, 2023, 22 pages. cited by applicant
Office Action received for European Patent Application No. 20172237.8 dated Jan. 22, 2024, 5 pages. cited by applicant
Office Action received for European Patent Application No. 20172238.6 dated Jan. 16, 2024, 5 pages. cited by applicant

---

Primary Examiner: Nguyen; Anh Tuan T

Assistant Examiner: Boler; Rynae E

Attorney, Agent or Firm: Faegre Drinker Biddle & Reath LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) The present application claims priority to and the benefit of European Patent Application Nos. 20172237.8 and 20172238.6, filed Apr. 30, 2020, which are incorporated by reference herein in their entirety.
TECHNICAL FIELD
(2) The present disclosure relates to insertable medical vision devices, such as, but not limited to, endoscopes, in particular disposable insertion endoscopes, such as duodenoscopes, gastroscopes, and colonoscopes. More specifically, the present disclosure relates to endoscope control systems comprising control wheels connected to associated wire drums for connection to steering wires, whereby rotation of the control wheels controls a bending operation of a tip of the endoscope.
BACKGROUND
(3) Endoscopes are typically equipped with a light source and a vision receptor including a vision or image sensor. Provided that enough light is present, it is possible for the operator to see where the endoscope is steered and to set the target of interest once the tip has been advanced thereto.
(4) Endoscopes typically comprise an elongated insertion tube with a handle at the proximal end, as seen from the operator, and visual inspection means, such as a built-in camera including a vision sensor, at a distal end of the elongated insertion tube. This definition of the terms distal and proximal, i.e. “proximal” being the end closest to the operator and “distal” being the end remote from the operator, as used herein for endoscopes in general, is adhered to in the present specification. Electrical wiring for the camera and other electronics, such as one or more LEDs accommodated in the tip part at the distal end, runs along the inside of the elongated insertion tube from the handle to the tip part. A working or suction channel may run along the inside of the insertion tube from the handle to the tip part, e.g. allowing liquid to be removed from the body cavity or allowing for insertion of surgical instruments or the like into the body cavity. The suction channel may be connected to a suction connector, typically positioned at a handle at the proximal end of the insertion tube.
(5) To be able to maneuver the endoscope inside the body cavity, the distal end of some endoscopes comprises a bendable distal tip, which may be bendable in one, e.g. an up/down dimension, or two dimensions, e.g. an up/down dimension and a left/right dimension. The bendable tip often comprises a bending section with increased flexibility, e.g. achieved by articulated segments of the bending section. The maneuvering of the endoscope inside the body is typically done by tensioning

or slacking steering wires also running along the inside of the elongated insertion tube from the tip part through the remainder of articulated segments to a control system or control mechanism positioned in or forming part of the handle.
(6) An endoscope control system for performing a bending operation in two dimensions is known from WO2018022418A2. This control system includes two control wheels connected to associated wire drums for connection to associated steering wires of the endoscope, whereby rotation of the control wheels controls the bending operation in two dimensions.
(7) U.S. Pat. No. 4,461,282 discloses another endoscope control system including two control wheels.

SUMMARY OF EMBODIMENTS OF THE DISCLOSURE

(8) A first aspect of the present disclosure relates to an endoscope control system for performing a bending operation in a disposable insertion endoscope, the endoscope control system comprising: an endoscope handle with a handle housing; a first control wheel connected to a first wire drum for connection to a first steering wire of the endoscope, whereby rotation of the first control wheel relative to the handle housing about an axis of rotation controls the bending operation in a first dimension, the first control wheel comprising a bearing surface; a second control wheel connected to a second wire drum for connection to a second steering wire of the endoscope, whereby rotation of the second control wheel relative to the handle housing about the axis of rotation controls the bending operation in a second dimension, the second control wheel comprising an outer bearing surface positioned farther from the axis of rotation than the bearing surface of the first control wheel; and an outer bearing element forming part of or being rotationally fixed to the handle housing, the outer bearing element comprising an inner bearing surface positioned farther from the axis of rotation than the outer bearing surface of the second control wheel, the inner bearing surface of the outer bearing element abutting the outer bearing surface of the second control wheel so that rotation of the second control wheel is at least partly borne on the outer bearing element.
(9) When the outer bearing element of the endoscope control system is provided on an outside of the second control wheel, a wall thickness of the outer bearing element may be less or not at all decisive for a total cross-sectional extent of the part of the control system, which does not include the handle. Therefore, it may be possible to allow this wall thickness to be larger, allowing for manufacture of the outer bearing element in a cheaper and/or less rigid and/or more environmentally friendly material, such as a plastic polymer, such as polyoxymethylene (POM). When the below described inner bearing element is positioned between the inner and outer control wheels, this may not apply to the inner bearing element described below, which may make it advantageous to manufacture the inner bearing element of a more rigid material and with a smaller wall thickness, such as polycarbonate (PC). When the second control wheel is borne on the outer bearing element, and thus not on an inner bearing element, less force is exerted on the inner bearing element, which may allow for the use of an inner bearing element (for which the cross-sectional extent may be more important) of a less rigid material and/or of smaller wall thickness, potentially resulting in a smaller cross-sectional extent or diameter. When the outer bearing element can be manufactured in a cheaper and/or less rigid and/or more environmentally friendly material, such as a plastic polymer, it may become advantageous to manufacture the outer bearing element in one piece with and/or of the same material as the handle housing, potentially saving material costs and/or weight and/or easing manufacture.
(10) The endoscope control system can alternatively be denoted an endoscope bending operation apparatus.
(11) The control system may be positioned on or in, or may form part, of an endoscope handle of the endoscope, see also further below.
(12) The first and/or second control wheels and/or the outer bearing element and/or the first and/or second wire drums may form part of the endoscope handle. The same is the case for the inner bearing element described below.

(13) The first control wheel may be positioned coaxially with the second control wheel and/or with the outer bearing element and/or the inner bearing element and/or the first and/or second wire drums.
(14) The handle housing may be or comprise a handle frame and/or may be manufactured of a plastic material, potentially a plastic polymer material and/or an artificial resin. One or both control wheels or at least the sleeves thereof described below, and/or one or both wire drums may be manufactured from one or more plastics or a plastics material, such as POM. The entire control system may be manufactured of one or more plastics or plastic materials or plastic polymer materials or artificial resins, and/or the control system does not include any metal. Any one of the plastic materials mentioned herein may be a plastic polymer material which comprises or consists of one or more of PC, polypropylene (PP), acrylonitrile butadiene styrene (ABS), polyethylene (PE), polyamide (PA), polyurethane (PU), polystyrene (PS), polylactic acid (PLA), polyvinyl chloride (PVC), polyoxymethylene (POM), polyester, polyethylene terephthalate (PET), and acrylic (PMMA). The polymer may be a copolymer of one or more monomers of the latter materials.
(15) Rotation of the control wheels may occur relative to the frame. The handle housing or the frame may be a handle shell or a housing shell. The frame and/or the housing may be manufactured of a rigid material, such as a rigid plastic polymer. The handle housing may be a handle shell.
(16) The first and/or second wire drums may be a pulley/pulleys. The first and second steering wires may form part of the control system and may be attached to the first and the second drum, respectively, and/or may be wound up or woundable on these, respectively.
(17) The axis of rotation may be an axis of rotation of also the first and/or second wheel sleeves described below.
(18) The axis of rotation may be a center axis of the first and/or second control wheel and/or of the first and/or second wheel handles described below and/or of the first and/or second wheel sleeves as described below and/or of the control system and/or of the outer bearing element and/or of the inner bearing element described below and/or of the center shaft described below.
(19) The first and second control wheels may comprise first and second control handles, respectively, of which one or both may comprise finger depressions. The second wheel handle may be positioned between the handle housing and the first wheel handle. A first wheel sleeve connecting the first wheel handle to the first wire drum may extend through, potentially through center holes of, the second wheel handle and/or through a second wheel sleeve, which connects the second wheel handle to the second wire drum, and/or through, potentially through a center opening of, the second wire drum.
(20) The second wire drum may be positioned between the first wire drum and the second control wheel.
(21) The first control wheel may be an inner control wheel and/or may be positioned closer to the handle housing than the second control wheel and/or may control a bendable tip of the endoscope in an up/down dimension. The second control wheel may be an outer control wheel and/or may control a bendable tip of the endoscope in a left/right dimension.
(22) The first control wheel may comprise a first wheel sleeve and potentially a first wheel handle, the first wheel sleeve connecting the first wheel handle to the first wire drum, the first wheel sleeve comprising the bearing surface of the first control wheel. Similarly, the second control wheel may comprise a second wheel sleeve and potentially a second wheel handle, the second wheel sleeve connecting the second wheel handle to the second wire drum, the second wheel sleeve comprising the outer bearing surface of the second control wheel. The second wheel sleeve may encompass at least part of the first wheel sleeve. The first and/or second wheel sleeves may each be cylindrical or circular cylindrical and/or may be embodied as cylinder shells and/or may be hollow. The second wheel sleeve may be positioned to be coaxial with the first wheel sleeve. The inner bearing element may be an inner bearing sleeve and may be the only element positioned between the first wheel

sleeve and the second wheel sleeve.
(23) The outer bearing element may be of the same material as or of a material different from, potentially of higher rigidity and/or stiffness, such as a metallic material, than that of the handle housing. The outer bearing element may be or include a potentially cylindrical outer bearing sleeve having a wall or a cylinder shell with a wall thickness of less than one fifth, one sixth, one seventh, one eighth, one ninth, or one tenth of a diameter or a cross-sectional extent of the outer bearing sleeve. The wall thickness of the outer bearing sleeve may be more than one twentieth, one fifteenth, one twelfth, one tenth, one ninth, one eighth, or seventh, or one sixth of a diameter or a cross-sectional extent of the outer bearing sleeve.
(24) The control system may further comprise the inner bearing element forming part of or being rotationally fixed to the handle housing. The inner bearing element may be of a material different from, potentially of higher rigidity and/or stiffness, than that of the handle housing. The inner bearing element may be or include a potentially cylindrical inner bearing sleeve having a wall or a cylinder shell with a wall thickness of less than one twentieth, one thirtieth, one fortieth, or one fiftieth of a diameter or a cross-sectional extent of the outer bearing sleeve. The wall thickness of the inner bearing sleeve may be less than, potentially less than half of, a quarter of, or an eight of that of the outer bearing sleeve.
(25) The first and second control wheels may be connected to first and second wire drums for connection to a steering wire of the endoscope, whereby rotation of the control wheel controls the bending operation.
(26) The control system may, for one or both control wheels, comprise a brake system, which may include a brake handle, or a similar activation device or activation means, movement of which moves a brake between a braking position and a released position. Such movement may be a rotation, potentially about a rotation axis of the control wheel. In the braking position, rotation of the associated control wheel may be braked or hindered. In the releasing position, rotation of the associated control wheel may be allowed.
(27) Any one of or all of the above-mentioned elements of the control system, potentially except for steering wires and/or parts of the brake system may be manufactured from plastic polymer(s).
(28) The second control wheel may be positioned coaxially with and potentially axially shifted in relation to the first control wheel. A diameter or a cross-sectional dimension of the two control wheels may be different from each other, potentially so that an outer one of the two control wheels has a smaller diameter or smaller cross-sectional dimension.
(29) In an embodiment of the control systems according to the present disclosure, the inner bearing surface of the outer bearing element and the outer bearing surface of the second control wheel each extends circumferentially about the axis of rotation.
(30) In another embodiment, the inner bearing surface of the outer bearing element and the outer bearing surface of the second control wheel each includes at least two bearing surface parts, which are positioned at an axial distance from each other.
(31) Hereby, more stability of the rotational movement may be provided.
(32) The bearing surface parts may alternatively be denoted bearing interfaces.
(33) The at least two bearing surface parts may consist of two or three bearing surface parts.
(34) Two of the at least two bearing surface parts may, respectively, be positioned at upper and lower parts or ends of the second wheel sleeve and the outer bearing sleeve. A third bearing surface part or more bearing surface parts may be provided between the two bearing surface parts positioned at upper and lower parts or ends of the second wheel sleeve and the outer bearing sleeve.
(35) Similarly, the inner bearing surface of the inner bearing element and the outer bearing surface of the first control wheel may each include two bearing surface parts, which are positioned at an axial distance from each other. And, similarly, the two bearing surface parts may, respectively, also here be positioned at upper and lower parts or ends of the first wheel sleeve and the inner bearing sleeve.

(36) In another embodiment, the outer bearing element is in one piece with the handle housing.
(37) The outer bearing element may be integral with and/or molded in one piece with the handle housing.
(38) Alternatively, the outer bearing element is provided separately from and is fixed to the handle housing.
(39) In another embodiment, the first control wheel comprises a first wheel handle and a first wheel sleeve, the first wheel sleeve connecting the first wheel handle to the first wire drum, the first wheel sleeve comprising the bearing surface of the first control wheel, wherein the second control wheel comprises a second wheel handle and a second wheel sleeve, the second wheel sleeve connecting the second wheel handle to the second wire drum, the second wheel sleeve comprising the outer bearing surface of the second control wheel, and wherein the second wheel sleeve encompasses at least part of the first wheel sleeve.
(40) The first and/or second wheel sleeves may each be cylindrical or circular cylindrical and/or may be embodied as cylinder shells and/or may be hollow.
(41) The second wheel sleeve may be positioned to be coaxial with the first wheel sleeve.
(42) The inner bearing element or an inner bearing sleeve may be the only element positioned between the first wheel sleeve and the second wheel sleeve.
(43) In a development of the above embodiment, the outer bearing element is an outer bearing sleeve that encompasses at least part of the second wheel sleeve.
(44) Similarly, the inner bearing element may be or comprise an inner bearing sleeve that encompasses at least part of the first wheel sleeve.
(45) The outer bearing sleeve may encompass at least part of the second wheel sleeve, which may again encompass at least part of the inner bearing sleeve, which may again encompass at least part of the first wheel sleeve. The second wheel sleeve may be positioned between and be rotational relative to the inner and outer bearing sleeves, which may each be static relative to the handle housing. The second wheel sleeve may be positioned between and be rotational relative to the inner and outer bearing sleeves, the second wheel sleeve being rotationally borne or supported on the outer bearing sleeve. The first wheel sleeve may be positioned on an interior side of the inner bearing sleeve and similarly be rotational relative to the inner and outer bearing sleeves, which may both be static relative to the handle housing.
(46) The outer bearing element may alternatively comprise an outer bearing sleeve that encompasses at least part of the second wheel sleeve.
(47) Another embodiment further comprises an inner bearing element forming part of or being rotationally fixed to the handle housing, the inner bearing element comprising an inner bearing surface positioned farther from the axis of rotation than the outer bearing surface of the second control wheel, the inner bearing surface of the bearing element abutting the outer bearing surface of the control wheel so that rotation of the second control wheel is at least partly borne on the inner bearing element.
(48) The second control wheel or a second wheel sleeve thereof may also be rotationally borne or supported on the inner bearing element, which may be or comprise an inner bearing sleeve, the inner bearing element comprising an outer bearing surface positioned closer to the axis of rotation than an inner bearing surface of the first control wheel, the outer bearing surface of the inner bearing element abutting an inner bearing surface of the second control wheel so that rotation of the first control wheel is at least partly borne on the inner bearing element.
(49) The first control wheel or a first wheel sleeve thereof may be rotationally borne or supported on the inner bearing element, which may be or comprise an inner bearing sleeve, the inner bearing element comprising an inner bearing surface positioned farther from the axis of rotation than an outer bearing surface of the first control wheel, the inner bearing surface of the inner bearing element abutting an outer bearing surface of the first control wheel so that rotation of the first control wheel is at least partly borne on the inner bearing element.

(50) Alternatively, the control system further comprises a center shaft, the first control wheel or a first wheel sleeve thereof instead, or in addition, being rotationally borne or supported on the center shaft, the center shaft forming part of or being rotationally fixed to the handle housing, the center shaft comprising an outer bearing surface positioned closer to the axis of rotation than an inner bearing surface of the first control wheel or a first wheel sleeve thereof, the outer bearing surface of the center shaft abutting the inner bearing surface of the first control wheel so that rotation of the first control wheel is at least partly borne on the center shaft. the first wheel sleeve may encompass at least part of the center shaft.
(51) In the embodiment where the first control wheel is borne on the inner bearing element, the center shaft may also be provided.
(52) The inner bearing element may generally be fixed to the, or to other parts of the, handle housing by means of screws and/or a snap connection.
(53) In another embodiment, the inner bearing element separates at least part of the first and second control wheels from each other so that rotation of the control wheels is mutually separated from each other.
(54) The inner bearing element may separate at least part of the first and second wheel sleeves from each other.
(55) Alternatively, or in addition, the two control wheels, or wheel sleeves thereof, may be located at a distance from each other and/or not be in contact with each other.
(56) In another embodiment, the inner bearing element is provided separately from and is fixed to the handle housing.
(57) Alternatively, the inner bearing element is in one piece with and/or integral with, and/or molded in one piece with the handle housing.
(58) In another embodiment, the outer bearing element is an outer bearing sleeve that projects from a surface of the handle housing toward a wheel handle of the second control wheel.
(59) The outer bearing sleeve may be a hollow cylinder.
(60) Alternatively, or in addition, the inner bearing element may be or comprise an inner bearing sleeve, which may also be a hollow cylinder, that projects from a surface of the handle housing toward a wheel handle of the second control wheel.
(61) In another embodiment, the second control wheel comprises a second wheel handle and a second wheel sleeve, the second wheel sleeve connecting the second wheel handle to the second wire drum, the second wheel sleeve comprising an inner part in one piece with the second wheel handle, the inner part of the second wheel sleeve being fixed to an outer part of the second wheel sleeve, the outer part of the second wheel sleeve being in one piece with the second wire drum, and wherein the outer bearing surface of the second control wheel is an outer surface of the outer part of the of the second wheel sleeve.
(62) Alternatively, or in addition, the first control wheel comprises a first wheel handle and a first wheel sleeve, the first wheel sleeve connecting the first wheel handle to the first wire drum, the first wheel sleeve comprising an outer part in one piece with the first wheel handle, the outer part of the first wheel sleeve being fixed to an inner part of the first wheel sleeve, the inner part of the first wheel sleeve being in one piece with the first wire drum, and wherein, potentially, the outer bearing surface of the first control wheel is an outer surface of the inner part of the of the first wheel sleeve.
(63) In another embodiment, the outer bearing element comprises a stop surface associated with a stop surface of the second control wheel so that when the second control wheel is rotated in a direction of rotation to an end position, mutual abutment of the stop surfaces prevents the second control wheel from rotating farther in that direction of rotation.
(64) The outer bearing element may comprise a further stop surface associated with a further stop surface of the second control wheel so that when the second control wheel is rotated in a direction of rotation opposite to the one defined above to an opposite end position, mutual abutment of the

further stop surfaces prevents the second control wheel from rotating farther in that opposite direction of rotation.

(65) The inner bearing element may similarly comprise a stop surface associated with a stop surface of the first control wheel so that when the first control wheel is rotated in a direction of rotation to an end position, mutual abutment of the stop surfaces prevents the first control wheel from rotating farther in that direction of rotation. The inner bearing element may similarly comprise a further stop surface associated with a further stop surface of the first control wheel so that when the first control wheel is rotated in a direction of rotation opposite to the latter one to an opposite end position, mutual abutment of the further stop surfaces prevents the first control wheel from rotating farther in that opposite direction of rotation.

(66) Hereby, restriction of rotation of one or both control wheels may achieved, which may accordingly restrict bending of a tip of the endoscope between bended end positions in the first or both the first and the second dimensions.

(67) In another embodiment, the outer bearing element comprises an outer bearing sleeve having a wall with a wall thickness of more than one twentieth of a diameter or a cross-sectional extent of the outer bearing sleeve, and wherein the control system further comprises an inner bearing element forming part of or being rotationally fixed to the handle housing, the inner bearing element comprising an inner bearing sleeve having a wall with a wall thickness of less than one thirtieth of a diameter or a cross-sectional extent of the outer bearing sleeve.

(68) In another aspect, the present disclosure involves an endoscope comprising a control system according to any one of the above embodiments and/or options.

(69) The endoscope may further comprise an endoscope handle at the proximal end thereof, and/or visual inspection means, such as a built-in camera including a vision sensor, at a distal tip. Electrical wiring for the camera and other electronics, such as one or more LEDs accommodated in the tip part at the distal end, may run along the inside of the elongated insertion tube from the endoscope handle to a PCB or an FPC at the distal tip. A working or suction channel may run along the inside of the insertion tube from the handle to the tip part, e.g. allowing liquid to be removed from the body cavity or allowing for insertion of surgical and/or sampling instruments or the like into the body cavity. The suction channel may be connected to a suction connector, typically positioned at a handle at the proximal end of the insertion tube.

(70) In some embodiments of the endoscope, the endoscope further comprises a distal tip or tip part that comprises a bending section connected to the steering wire(s) so that the control system can activate a bending operation of the bending section via the steering wire(s).

(71) The bending section may be bendable in one or two dimensions, e.g. an up/down dimension and a left/right dimension. The bendable tip may comprise a bending section with increased flexibility, e.g. achieved by articulated segments of the bending section as are known in the art. The steering wire(s) may run along the inside of an elongated insertion tube from the tip through the bending section to the control system positioned in or forming part of the endoscope handle.

(72) The endoscope may be a disposable insertion endoscope. The endoscope may include one or more features as described herein in the above, including the features of endoscopes described in the above introduction to this description, and in connection with the description of the methods and tip parts according to the present disclosure.

(73) The endoscope may comprise an elongated insertion tube with a handle at the proximal end. A tip or tip part may be positioned at the distal end of the elongated insertion tube. The tip may further comprise a bending section positioned between the tip and the elongated insertion tube. The bending section may be configured to be articulated to maneuver the endoscope inside a body cavity.

(74) The endoscope may be a duodenoscope, a gastroscope, or a colonoscope.

(75) An embodiment of the endoscope further comprises the first and second steering wires and a distal tip or tip part that comprises a bending section connected to the first and second steering

wires so that the control system can activate the bending operation of the bending section via the steering wires.
(76) A person skilled in the art will appreciate that any one or more of the above aspects of this disclosure and embodiments thereof may be combined with any one or more of the other aspects and embodiments thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the following, non-limiting exemplary embodiments will be described in greater detail with reference to the drawings, in which:
(2) FIG. **1** shows a perspective view of an endoscope including control system according to the present disclosure;
(3) FIG. **2** shows a top view of the control system of FIG. **1**;
(4) FIG. **3** shows a cross-sectional view taken along line III-III in FIG. **2**;
(5) FIG. **4** shows an exploded perspective view of a handle frame and the control system of the endoscope of FIG. **1**;
(6) FIG. **5** shows an exploded side view of a first control wheel and a first shaft unit of the control system of FIG. **1**;
(7) FIG. **6** shows an exploded side view of the first control wheel and first shaft unit of FIG. **5** in an assembled state and in which the first control wheel and first shaft unit have been turned 180 degrees;
(8) FIG. **7** shows a cross section taken along the line VII-VII in FIG. **6**;
(9) FIG. **8** shows an exploded side view of a second control wheel and a second shaft unit of the control system of FIG. **1**;
(10) FIG. **9** shows a cross-sectional view taken along the line IX-IX of FIG. **8**; and
(11) FIG. **10** shows a view like that of FIG. **9**, wherein the second control wheel and second shaft unit are in an assembled state.
DETAILED DESCRIPTION
(12) In this disclosure, the term “to accommodate” may additionally or alternatively be defined as “to house” or “to enclose” or “to surround”.
(13) In this specification, the terms “integrally” or “integrally provided” or “integrally comprising”, “in one piece”, or similar may be defined as the associated features forming an integral part of a whole; and/or are in one piece, potentially molded in one piece; and/or are substantially inseparable by hand. When a first element forms part of a second element, this may involve that the first element is provided integrally with or in one piece with the second element.
(14) As mentioned, in this specification, the term “proximal” may be defined as being closest to an operator of the endoscope, and the term “distal” as being remote from the operator. The term “proximal-distal” may be defined as extending between these two extremes, in the present case proximal-distal may extend along a center axis of the tip part extending between a proximal extremity of the proximal end of the tip part and a distal extremity of the distal end of the tip part.
(15) In this specification, an endoscope may be defined as a device adapted for viewing body cavities and/or channels of a human and/or animal body. The endoscope may for instance be a flexible or steerable endoscope. The endoscope may be a duodenoscope or a ureteroscope, a gastroscope, or a colonoscope.
(16) FIG. **1** shows a disposable insertion endoscope **1** with a control system **100**, an elongated insertion tube **3**, and an endoscope handle **2** at a proximal end **3***a* of the elongated insertion tube **3**. The endoscope handle includes a handle housing **116**
(17) In a known manner, an endoscope tip **4** is positioned at a distal end **3***b* of the elongated

insertion tube **3**, the tip **4** comprising a bending section **5** positioned between the tip **4** and the elongated insertion tube **3**. The endoscope handle **2** comprises the endoscope control system **100**, the endoscope control system **100** being for performing a bending operation of the disposable insertion endoscope **1**.

(18) In a known manner, the bending section **5** is connected to steering wires **102***k*, **102***l*, which extend from the control system **100** through the tube **3** to allow the control system **100** to activate a two-dimensional bending operation of the bending section **5** via the steering wires. The bending section **5** is configured to be articulated to maneuver the endoscope **1** inside a body cavity (not shown). The bending section **5** is bendable in two dimensions, i.e. an up/down dimension and a left/right dimension.

(19) The bending section **5** has increased flexibility achieved by articulated segments of the bending section **5** as is known in the art. The steering wires run along the inside of the elongated insertion tube **3** from the tip **4** through the bending section **5** to the endoscope control system **100**. Still in a known manner, the maneuvering of the endoscope **1** inside the body can be carried out by tensioning or slacking the steering wires by means of the control system **100**.

(20) Still in a known manner, the distal tip **4** has a not shown built-in camera including a vision sensor. Not shown electrical wiring for the camera and potential other electronics, such as one or more LEDs accommodated in the tip part **4**, run along the inside of the elongated insertion tube **3** from the endoscope handle **2** to a PCB or an FPC at or in the distal tip **4**. A not shown suction/working channel runs along the inside of the insertion tube **3** from the handle **2** to the tip part **4**, e.g. allowing liquid to be removed from the body cavity or allowing for insertion of a surgical instrument and/or a sampling instrument or other instruments (not shown) into the body cavity. The suction channel is connected to a suction connector **6** positioned at the proximal end of the handle **2**. A sampling connector **81** is positioned at the distal end of the handle **2**.

(21) Referring to FIGS. **1** to **7**, the control system **100** comprises a first control wheel **101** connected to a first wire drum **102***a* for connection to a steering wire of the endoscope **1**, whereby rotation of the first control wheel **101** controls the bending operation in one dimension by rotating the wire drum **102***a* to, in a known manner, activate the steering wire, the steering wire being connected to the wire drum **102***a*.

(22) The first control wheel **101** comprises a first shaft/sleeve **102***e* which connects a first wheel handle **101***c* of the first control wheel **101** to the first wire drum **102***a* for connection to a first steering wire **102***k* of the endoscope **1**. The first shaft **102***e* comprises an outer bearing surface **102***i*. The first shaft **102***e* and the first wire drum **102***a* are provided as a one-piece element denoted a first shaft unit **102***c*. The first wheel handle **101***c* and the first central part **102***g* are provided as a one-piece element denoted a first control wheel unit **101***a*. Hereby, rotation of the first control wheel **101** relative to the handle housing **116** about an axis of rotation controls the bending operation in the first dimension.

(23) Referring to FIGS. **1** to **4** and **8** to **10**, the control system **100** also comprises a second control wheel **102** similarly connected to a second wire drum **102***b* for connection to another steering wire of the endoscope **1**, whereby rotation of the second control wheel **102** controls the bending operation in another dimension by rotating the wire drum **102***b* to activate the steering wire, the steering wire being connected to the wire drum **102***b*.

(24) The second control wheel **102** comprises a second shaft/sleeve **102***f* which connects a second wheel handle **101***d* of the second control wheel **102** to the second wire drum **102***b* for connection to a second steering wire **102***l* of the endoscope **1**. The second shaft **102***f* and the second wire drum **102***b* are provided as a one-piece element denoted a second shaft unit **102***d*. The second shaft/sleeve **102***f* comprises an outer bearing surface **102***j*. The second wheel handle **101***d* and the second central part **102***h* are provided as a one-piece element denoted a second control wheel unit **101***b*. Hereby, rotation of the second control wheel **102** relative to the handle housing **116** about the axis of rotation controls the bending operation in a second dimension. The outer bearing surface

102*j* of the second control wheel **102** is positioned farther from the axis of rotation than the bearing surface **102***i* of the first control wheel **101**.

(25) An outer bearing element **120***b* is in one piece with a housing frame **116***a* of the handle housing **116**, the outer bearing element **120***b* comprising an inner bearing surface **120***d* positioned farther from the axis of rotation than an outer bearing surface **102***j* of the second control wheel **102**, the inner bearing surface **120***d* of the outer bearing element **120***b* abutting the outer bearing surface **102***j* of the second control wheel **102** so that rotation of the second control wheel **102** is at least partly borne on the outer bearing element **120***b*. The outer bearing element **120***b* is shaped as a cylindrical sleeve having a wall or a cylinder shell with a wall thickness.

(26) As shown in FIG. **1**, the control system **100** is partly positioned in the handle housing **116** to form part of the endoscope handle **2**.

(27) Rotation of the control wheels **101**, **102** occurs relative to a housing frame **116***a*, which is a half part of whole housing frame forming the handle housing **116** which when assembled as shown in FIG. **1** comprises another half part as well.

(28) The inner bearing surface **120***d* of the outer bearing element **120***b* and the outer bearing surface **102***j* of the second control wheel **102** each extends circumferentially about the axis of rotation. The inner bearing surface **120***d* and the outer bearing surface **102***j* each includes two circumferentially extending bearing surface parts/portions or bearing interfaces **102***jbs* (see FIGS. **3** and **9**), which are positioned at an axial distance from each other. A non-bearing portion **102***jnbs* is shown between the bearing surface portions. In these interfaces, the surfaces **120***d* and **102***j* are in abutment with each other. Hereby, stability of the rotational movement is provided. The bearing surface parts are positioned at upper and lower parts or ends of the second shaft/sleeve **102***e* and the outer bearing element/sleeve **120***b*.

(29) Similarly, the inner bearing surface **120***c* of the inner bearing element **120***a* and the outer bearing surface **102***i* of the first control wheel **101** each includes two bearing surface parts, which are similarly positioned at an axial distance from each other.

(30) The outer bearing element **120***b* is in one piece with the handle frame **116***a* of the handle housing **116**.

(31) The first and second shafts/sleeves are each cylindrical or, rather, slightly conical, and hollow.

(32) The second shaft/sleeve **102***f* encompasses a part of the first shaft/sleeve **102***e*. The outer bearing element **120***b* encompasses part of the second shaft/sleeve **102***f*. The inner bearing element **120***a* comprises an inner bearing sleeve or sleeve part **120***e* that encompasses part of the first shaft/sleeve **102***e*. The outer bearing element encompasses part of the second shaft/sleeve **102***f*, which again encompasses part of the sleeve part **120***e*, which again encompasses part of the first shaft/sleeve **102***e*.

(33) The second shaft/sleeve **102***f* is positioned between and is rotational relative to the inner and outer bearing elements **120***a*, **120***b*, which are static relative to the handle housing **116**, the second shaft/sleeve **102***f* being rotationally borne or supported on the outer bearing element **120***b*. The first shaft/sleeve **102***e* is positioned on an interior side of the inner bearing element **120***a* and is rotational relative to the inner and outer bearing elements **120***a*, **120***b*, which are both static relative to the handle housing **116**.

(34) The inner bearing element **120***a* separates the first and second control wheels **101**, **102** from each other so that rotation is mutually separated.

(35) The first shaft/sleeve **102***e* embodies the first wheel sleeve mentioned above, and the second shaft/sleeve **102***f* embodies the second wheel sleeve mentioned above.

(36) The first and second wire drums **102***a*, **102***b* are positioned inside the assembled handle housing **116**.

(37) The first and second wheel handles **101***c*, **101***d* are generally circular and comprise conventionally provided finger depressions or cut-outs.

(38) Each of the first and second control wheels **101**, **102** comprises a central part **102***g*, **102***h*, first

and second, respectively, each surrounding a center opening. The central parts **102***g*, **102***h* are cylindrical and extend towards the housing frame **106***a*. The second central part **102***h* may in other embodiments extend to encompass part of the first central part **102***g* in the assembled state of the control system **100**.

(39) The first and second shafts **102***e*, **102***f* are each tubular and each comprises a substantially cylindrical or, rather, slightly conical circumferential wall which provide the associated bearing surfaces **102***i*, **102***j*. A diameter of the first shaft **102***e* is smaller than that of the second shaft **102***f*.

(40) The axes of rotation of the control wheels **101**, **102** are coinciding to form one axis of rotation, which is also a center axis of the control system **100**. This axis extends in an assembly direction D in which the different elements of the control system are assembled. The first and second shafts **102***e*, **102***f*, the first and second control wheels **101**, **102**, and the first and second wheel handles **101***c*, **101***d* extend coaxially in the assembled control system **100**.

(41) As shown in FIG. **4**, the first and second wire drums **102***a*, **102***b* on the one hand and the first and second wheel handles **101***c*, **101***d* on the other hand are positioned on opposite sides of the connection hole **116***b* of the housing frame **116***a*. The first wire drum **102***a* is positioned in extension of the second wire drum **102***b* and farther from the housing frame **116***a* or the connection hole **116***b* than the second wire drum **102***b*.

(42) The first and second wire drums **102***a*, **102***b* are positioned at upper ends of the first and second shafts **102***e*, **102***f*, respectively.

(43) The control system **100** further includes a center shaft **103** which extends through a connection hole **116***b* of the housing frame **116***a*, the first and second shaft units **102***c*, **102***d*, and the center openings of the first and second control wheels **101**, **102**. The center shaft **103** comprises a connector frame or center shaft frame **115**. The connector frame **115** is fixed to the housing frame **116***a* via the inner bearing element frame **121**, see below, after insertion of the center shaft **103**. Hereby, the center shaft **103** is fixed to the housing frame **116***a*. The connector frame **115** extends radially from a shaft part **103***a* of the center shaft **103** and is positioned within the handle housing **116***a* in the assembled endoscope **1**, see FIG. **3**. The connector frame **115** includes a housing **115***a* including a flange **115***b* fixed to the housing frame. The housing **115***a* includes an indentation **115***c* that operates as stop or stop surface when it contacts a corresponding stop surface **102***m* on a longitudinally protruding portion of the first shaft unit **102***c* as it rotates. The protruding portion has an arcuate shape and its length determines the angle of rotation of the first shaft unit **102***c*.

(44) Similarly, the inner bearing element **120***a* includes an inner bearing element frame **121** that extends radially from a sleeve part **120***e* thereof and is positioned within the handle housing **116** in the assembled endoscope **1**. The inner bearing element frame **121** is directly fixed to the housing frame **116***a*, the center shaft frame **115** being directly fixed to the inner bearing element frame **121** by means of screws (not shown) so as to be indirectly fixed to the housing frame **116***a*. The screws are inserted into screw holes, one of these being designated **122** in FIG. **4**.

(45) The first shaft unit **102***c* is snapped into engagement with the first control wheel unit **101***a* by means of a first snap connection **112** between the first shaft **102***e* and the first control wheel unit **101***a*. The first snap connection **112** includes two connection parts **112***a* of the first shaft unit **102***c* interlocking with two associated connection parts **112***b* (best seen in FIG. **7**) of the first control wheel unit **101***a*. These associated connection parts **112***a*, **112***b* mutually engage. During the movement of the first shaft unit **102***c*, the connection parts **112***a* are pushed inwardly in the radial direction by the wheel sleeve **102***g* and, then, when the first shaft unit **102***c* is further moved and inserted, resiliently snap back to engage the associated connection part **112***b*. The connection parts **112***b* include a ramp or inclined surface **112***c*, which forces the pushable connection parts **112***a* in the radial direction during the movement of the first shaft unit **102***c*. Accordingly, the pushable connection parts **112***a* include a barb surface **112***d* which during the snap moves into engagement with an associated, opposed barb surface **112***e* of the connection parts **112***b* to secure the position of the first shaft unit **102***c* to the first control wheel unit **101***a*.

(46) The second shaft unit **102***d* is snapped into engagement with the second control wheel unit **101***b* by means of a second snap connection **113** between the second shaft **102***f* and the second control wheel unit **101***b*. The second snap connection **103** includes two connection parts **113***a* in the form of resilient projections or pins of the second control wheel unit **101***b* which interlock with two associated connection parts **113***b*, embodied by recesses, of the second shaft unit **102***d*. These connection parts **113***a*, **113***b* are thus mutually engaging. The connection parts **113***b* includes a ramp or inclined surface **113***c*, which forces the pushable connection parts **113***a* inwardly in a radial direction during the movement of the second shaft unit **102***d*. Accordingly, the pushable connection parts **113***a* include a barb surface **113***d* which during the snap moves into engagement with an associated barb surface **113***e* of the other connection parts **113***b* to secure the position of the second shaft unit **102***d* to the second control wheel unit **101***b*.
(47) A cap **105***a* is attached to a tip end **103***b* of the center shaft **103** by another snap connection **103***c* which is provided in a manner similar to the first and second snap connections **112**, **113**. Accordingly, the cap **105***a* includes two resilient and pushable connection parts **103***d*, whereas the tip end **103***b* includes associated two connection parts taking the form of recesses **103***e*. This snap engagement **103***c* is similarly be activated during or at the end of the insertion of the center shaft **103** into the control system **100**.
(48) The cap **105***a* covers and attaches a first multi-disc brake **110***a* of the first control wheel **101**, see further below. The brake **110***a* is encased within a spacing defined by interior surfaces of the first wheel handle **101***c*. The cap **105***a* includes a brake knob **104***a* projecting in the assembly direction and upon rotation of which the brake **110***a* is activated to brake rotation of the first control wheel **101** and, thus, first shaft unit **102***d*.
(49) A brake handle **104***b* for activation of a similar, second multi-disc brake **110***b*, which brakes the second control wheel **102** in a similar manner, is attached to the housing frame **116***a*.
(50) As shown in FIG. **3**, the first and second control wheels **101**, **102** house the associated brakes **110***a*, **110***b*. Activation of each of the brakes **110***a*, **110***b* moves the brake from a released position to a braking position. A brake force of the brake **110***a*, **110***b* in the braking position brakes rotation of the associated control wheel **101**, **102**, respectively. The brake force is released in the released position. The brakes **110***a*, **110***b* each includes a stack of brake discs and a helical spring **117***a*, **117***b*, respectively.
(51) The first and/or second wire drums **102***a*, **102***b* are pulleys. The first and second steering wires are attached to the wire drums **102***a*, **102***b* to be woundable on these, respectively.
(52) The foregoing aspects are further embodied in the following exemplary items:
(53) Item 1. An endoscope control system for performing a bending operation in a disposable insertion endoscope, the endoscope control system comprising: an endoscope handle with a handle housing; a first control wheel connected to a first wire drum for connection to a first steering wire of the endoscope, whereby rotation of the first control wheel relative to the handle housing about an axis of rotation controls the bending operation in a first dimension, the first control wheel comprising a bearing surface; a second control wheel connected to a second wire drum for connection to a second steering wire of the endoscope, whereby rotation of the second control wheel relative to the handle housing about the axis of rotation controls the bending operation in a second dimension, the second control wheel comprising an outer bearing surface positioned farther from the axis of rotation than the bearing surface of the first control wheel; and an outer bearing element forming part of or being rotationally fixed to the handle housing, the outer bearing element comprising an inner bearing surface positioned farther from the axis of rotation than the outer bearing surface of the second control wheel, the inner bearing surface of the outer bearing element abutting the outer bearing surface of the second control wheel so that rotation of the second control wheel is at least partly borne on the outer bearing element.
(54) Item 2. The control system according to item 1, wherein the inner bearing surface of the outer bearing element and the outer bearing surface of the second control wheel each extends

circumferentially about the axis of rotation.
(55) Item 3. The control system according to item 1 or 2, wherein the inner bearing surface of the outer bearing element and the outer bearing surface of the second control wheel each includes at least two bearing surface parts, which are positioned at an axial distance from each other.
(56) Item 4. The control system according to any one of the previous items, wherein the outer bearing element is in one piece with the handle housing.
(57) Item 5. The control system according to item 4, wherein the outer bearing element is molded in one piece with the handle housing.
(58) Item 6. The control system according to any one of the previous items, wherein the first control wheel comprises a first wheel handle and a first wheel sleeve, the first wheel sleeve connecting the first wheel handle to the first wire drum, the first wheel sleeve comprising the bearing surface of the first control wheel, wherein the second control wheel comprises a second wheel handle and a second wheel sleeve, the second wheel sleeve connecting the second wheel handle to the second wire drum, the second wheel sleeve comprising the outer bearing surface of the second control wheel, and wherein the second wheel sleeve encompasses at least part of the first wheel sleeve.
(59) Item 7. The control system according to item 6, wherein the outer bearing element is an outer bearing sleeve that encompasses at least part of the second wheel sleeve.
(60) Item 8. The control system according to any one of the previous items, further comprising an inner bearing element forming part of or being rotationally fixed to the handle housing, the inner bearing element comprising an inner bearing surface positioned farther from the axis of rotation than the outer bearing surface of the second control wheel, the inner bearing surface of the bearing element abutting the outer bearing surface of the control wheel so that rotation of the second control wheel is at least partly borne on the inner bearing element.
(61) Item 9. The control system according to item 8, wherein the inner bearing element separates at least part of the first and second control wheels from each other so that rotation of the control wheels is mutually separated from each other.
(62) Item 10. The control system according to item 8 or 9, wherein the inner bearing element is provided separately from and is fixed to the handle housing.
(63) Item 11. The control system according to any one of the previous items, wherein the outer bearing element is an outer bearing sleeve that projects from a surface of the handle housing toward a wheel handle of the second control wheel.
(64) Item 12. The control system according to any one of the previous items, wherein the second control wheel comprises a second wheel handle and a second wheel sleeve, the second wheel sleeve connecting the second wheel handle to the second wire drum, the second wheel sleeve comprising an inner part in one piece with the second wheel handle, the inner part of the second wheel sleeve being fixed to an outer part of the second wheel sleeve, the outer part of the second wheel sleeve being in one piece with the second wire drum, and wherein the outer bearing surface of the second control wheel is an outer surface of the outer part of the of the second wheel sleeve.
(65) Item 13. The control system according to any one of the previous items, wherein the outer bearing element comprises a stop surface associated with a stop surface of the second control wheel so that when the second control wheel is rotated in a direction of rotation to an end position, mutual abutment of the stop surfaces prevents the second control wheel from rotating farther in that direction of rotation.
(66) Item 14. The control system according to any one of the previous items, wherein the outer bearing element comprises an outer bearing sleeve having a wall with a wall thickness of more than one twentieth of a diameter or a cross-sectional extent of the outer bearing sleeve, and wherein the control system further comprises an inner bearing element forming part of or being rotationally fixed to the handle housing, the inner bearing element comprising an inner bearing sleeve having a wall with a wall thickness of less than one thirtieth of a diameter or a cross-sectional extent of the

outer bearing sleeve.

(67) Item 15. An endoscope comprising a control system according to any one of items 1 to 14.

(68) Item 16. The endoscope according to item 15, further comprising the first and second steering wires and a distal tip or tip part that comprises a bending section connected to the first and second steering wires so that the control system can activate the bending operation of the bending section via the steering wires.

LIST OF REFERENCE SIGNS

(69) **1** Endoscope **2** Endoscope handle **3** Elongated insertion tube **3***a* Proximal end of insertion tube **3** **3***b* Distal end of insertion tube **3** **4** Tip **5** Bending section **6** Suction connector **7** Working channel port **100** Endoscope control system **101** First control wheel **101***a* First control wheel unit **101***b* Second control wheel unit **101***c* First wheel handle **101***d* Second wheel handle **102** Second control wheel **102***a* First wire drum **102***b* Second wire drum **102***c* First shaft unit **102***d* Second shaft unit **102***e* First shaft/sleeve **102***f* Second shaft/sleeve **102***g* First central part **102***h* Second central part **102***i* Outer bearing surface of first control wheel **102***j* Outer bearing surface of second control wheel **103** Center shaft **103***a* Shaft part **103***c* Snap connection **103***d* Connection parts **103***e* Recesses **104***a* Brake knob **104***b* Brake handle **105***a* Cap **110***a* First multi-disc brake **110***b* Second multi-disc brake **111** Stack of brake discs **112** First snap connection **112***a* Snap connection parts **112***b* Snap connection parts **112***c* Ramps **112***d* Barb surface **112***e* Barb surface **113** Second snap connection **113***a* Snap connection parts **113***b* Snap connection parts **113***c* Ramps **113***d* Barb surface **113***e* Barb surface **116** Handle housing **111** Stack of brake discs **116***a* Housing frame **116***b* Connection hole **117***a* Spring **117***b* Spring **120***a* Inner bearing element **120***b* Outer bearing element **120***c* Inner bearing surface of inner bearing element **120***d* Inner bearing surface of outer bearing element **120***e* Sleeve part **121** Inner bearing element frame **122** Screw hole D Assembly direction

Claims

1. An endoscope comprising: an endoscope handle with a handle housing, the handle housing comprising two housing shells and an outer bearing element molded in one piece with one of the two housing shells; and a control system including: a first control wheel including a first shaft unit, a first control wheel unit and a first snap connection connecting the first shaft unit and the first control wheel unit, the first control wheel operable to rotate about an axis of rotation, the first shaft unit comprising a first wire drum, a distal end, a first wheel sleeve having a bearing surface, and two connection parts at the distal end in the form of distally extending flexible fingers, each of the flexible fingers comprising a radially outwardly extending barb, and the first control wheel unit comprising a first wheel handle and a first central part, the first central part comprising a cavity and two connection parts, each of the two connection parts of the first central part comprising a barb surface, the first snap connection formed when the distal end of the first shaft unit and the two connection parts of the first shaft unit are received in the cavity of the first central part with the radially outwardly extending barbs abutting the barb surfaces of the two connection parts of the first central part to prevent separation of the first control wheel unit from the first shaft unit; a second control wheel including a second shaft unit, a second wheel handle and a second snap connection connecting the second shaft unit and the second wheel handle, the second shaft unit comprising a second wire drum and a second wheel sleeve having an outer bearing surface positioned farther from the axis of rotation than the bearing surface of the first control wheel; and the outer bearing element comprising an inner bearing surface positioned farther from the axis of rotation than the outer bearing surface of the second control wheel, the inner bearing surface of the outer bearing element abutting the outer bearing surface of the second control wheel so that rotation of the second control wheel is at least partly borne on the outer bearing element, wherein the outer bearing surface of the second control wheel includes two bearing surface portions axially spaced apart from each other and a non-bearing surface portion between the two bearing surface

portions, and wherein the two bearing surface portions rotatably bear on the inner bearing surface of the outer bearing element.
2. The endoscope according to claim 1, wherein the inner bearing surface of the outer bearing element and the outer bearing surface of the second control wheel each extends circumferentially about the axis of rotation.
3. The endoscope according to claim 1, wherein the second wheel sleeve encompasses at least part of the first wheel sleeve.
4. The endoscope according to claim 3, wherein the outer bearing element comprises an outer bearing sleeve that encompasses at least part of the second wheel sleeve.
5. The endoscope according to claim 1, further comprising an inner bearing element forming part of or being rotationally fixed to the handle housing, the inner bearing element comprising an inner bearing surface positioned farther from the axis of rotation than the bearing surface of the first control wheel, the inner bearing surface of the inner bearing element abutting the bearing surface of the first control wheel so that rotation of the first control wheel is at least partly borne on the inner bearing element.
6. The endoscope according to claim 5, wherein the inner bearing element separates at least part of the first and second control wheels from each other so that rotation of the control wheels is mutually separated from each other.
7. The endoscope according to claim 5, wherein the inner bearing element is provided separately from and is fixed to the handle housing.
8. The endoscope according to claim 1, wherein the outer bearing element is an outer bearing sleeve that projects from a surface of the handle housing toward the second wheel handle.
9. The endoscope according to claim 1, wherein the second wheel sleeve includes an outer part opposite and in one piece with the second wire drum, the second wheel sleeve including the outer bearing surface between the outer part and the second wire drum, and the second control wheel comprising an inner part in one piece with the second wheel handle, the inner part being fixed to the outer part of the second wheel sleeve.
10. The endoscope according to claim 1, wherein the second control wheel comprises a stop surface, and wherein the outer bearing element comprises a stop surface associated with the stop surface of the second control wheel so that when the second control wheel is rotated in a direction of rotation to an end position defined by the stop surfaces, mutual abutment of the stop surfaces prevents the second control wheel from rotating farther in that direction of rotation.
11. The endoscope according to claim 1, further comprising an inner bearing element forming part of or being rotationally fixed to the handle housing, wherein the outer bearing element comprises an outer bearing sleeve having a wall with a wall thickness of more than one twentieth of a diameter or a cross-sectional extent of the outer bearing sleeve, and wherein the inner bearing element comprises an inner bearing sleeve having a wall with a wall thickness of less than one thirtieth of a diameter or a cross-sectional extent of the outer bearing sleeve.
12. The endoscope according to claim 1, further comprising a first steering wire and a second steering wire, and a distal tip or tip part that comprises a bending section connected to the first and second steering wires so that the control system can bend the bending section via the first and second steering wires.
13. The endoscope according to claim 12, wherein rotation of the first control wheel relative to the handle housing about the axis of rotation bends the bending section in a first dimension; and wherein rotation of the second control wheel relative to the handle housing about the axis of rotation bends the bending section in a second dimension.
14. The endoscope according to claim 5, further comprising a center shaft including a center shaft frame and a shaft part, wherein the inner bearing element includes an inner bearing element frame and a sleeve part having the inner bearing surface, wherein the center shaft frame is directly affixed to the inner bearing element frame, and wherein the shaft part extends through a connection hole in

the handle housing and through the first wheel sleeve.

15. The endoscope according to claim 1, further comprising an inner bearing element forming part of or being rotationally fixed to the handle housing, the inner bearing element comprising an inner bearing sleeve having a wall with a wall thickness and an inner bearing surface abutting the bearing surface of the first control wheel so that rotation of the first control wheel is at least partly borne on the inner bearing element, wherein the inner bearing sleeve is positioned between the first wheel sleeve and the second wheel sleeve, the second wheel sleeve and the inner bearing sleeve being sized and configured so that the inner bearing sleeve does not bear the rotation of the second control wheel, wherein the outer bearing element comprises an outer bearing sleeve.

16. The endoscope according to claim 15, wherein the outer bearing sleeve comprises a wall including the inner bearing surface and having a wall thickness is less than one-tenth of an outer diameter of the outer bearing sleeve.

17. The endoscope according to claim 16, wherein the wall thickness of the inner bearing sleeve is less than the wall thickness of the outer bearing sleeve.

18. The endoscope according to claim 15, wherein the inner bearing sleeve consists essentially of a polycarbonate polymer.

19. The endoscope according to claim 15, wherein the inner bearing element comprises an inner bearing element frame connected to the inner bearing sleeve and the handle housing, the inner bearing element surrounding the second wire drum.

20. The endoscope according to claim 1, further comprising a center shaft forming part of or being rotationally fixed to the handle housing, the first wheel sleeve receiving the center shaft with the bearing surface facing and abutting an outer bearing surface of the center shaft so that rotation of the first control wheel is at least partly borne on the center shaft, wherein the outer bearing element is molded in one piece with the handle housing and comprises an outer bearing sleeve having a wall including the inner bearing surface, and wherein the second wheel sleeve is positioned between the first wheel sleeve and the outer bearing sleeve.